Quantification: I534_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	29
interior lacuna count	19
frame-edge lacuna count	10

Interior lacunae only: mean, median and sample SD (n = 19)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2545.05	2059	1723.61	px^2
major_axis_length_px	lacuna shape	77.6596	74.1218	34.4772	px
minor_axis_length_px	lacuna shape	42.9122	43.4719	14.5489	px
aspect_ratio	lacuna shape	1.8145	1.6502	0.574	unitless
eccentricity	lacuna shape	0.762	0.7955	0.173	unitless
solidity	lacuna shape	0.8115	0.8447	0.105	unitless
roots_count	headline	7.5263	6	4.587	count
ring_length_r30_px	headline	240.316	234	116.152	px
ring_length_r60_px	reported	502.895	447	203.359	px
owned_length_px	ownership-dependent	512.488	386.25	401.375	px
edge_count	ownership-dependent	25.1053	21	19.7059	count
mean_edge_length_px	ownership-dependent	21.1716	20.5853	5.8078	px
perimeter_px	normalised	254.023	249.078	104.31	px
ring_area_r30_px2	normalised	8883	8451	2820.15	px^2
ring_area_r60_px2	normalised	19064.2	20074	5536.87	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9921	1	0.0223	unitless
ring_density_r30	normalised	0.0271862	0.0248108	0.0110759	px^-1
ring_density_r60	normalised	0.0265744	0.0257097	0.00753701	px^-1
roots_per_100px_perimeter	normalised	2.9348	2.6103	1.477	per 100 px
ring_attached_length_r30_px	network v2	206.158	201	119.467	px
ring_attached_length_r60_px	network v2	349.474	297	220.56	px
ring_length_w_r30_px	network v2	273.239	265.723	133.276	px
ring_length_w_r60_px	network v2	573.01	502.961	231.205	px
sholl_crossings_r10	network v2	5.6842	6	2.8098	count
sholl_crossings_r20	network v2	6.5789	6	3.7463	count
sholl_crossings_r30	network v2	7.3158	6	3.2327	count
loop_count	loops, report only	1.2632	1	2.0505	count
loop_edge_share	loops, report only	0.1178	0.0741	0.1431	share
ring_width_mean_r30_px	width	7.5405	7.5827	0.967	px

Per field
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measure	value	units
analysed_area_px2	914989	px^2
total_skeleton_length_px	23353	px
canalicular_length_density_per_px	0.0255227	px^-1
junction_count	457	count
junction_density_per_px2	0.00049946	px^-2
lacuna_count	29	count

lacunae_per_px2	3.169e-05	px^-2
skeleton_component_count	300	count
mean_component_length_px	77.8433	px
median_component_length_px	27	px
field_length_density_w_per_px	0.0290052	px^-1
field_density_without_flagged_per_px	0.0265537	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	268	px
unattached_component_count	18	count
canalicular_length_density_attached_per_px	0.0255227	px^-1
field_length_density_with_unattached_per_px	0.0258156	px^-1
loop_count	39	count
loop_edge_share	0.1076	share
edge_count	1218	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9528	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	22290	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6611	188.219	59.1114	3.1841	0.9494	0.7758	-1.5406	20.9109	224.092	True
2	1077	56.963	28.6202	1.9903	0.8646	0.7016	0.9022	20.0604	411.138	True
3	10729	252.088	59.2099	4.2575	0.972	0.8231	-1.467	29.4163	690.039	True
4	7272	192.335	53.4975	3.5952	0.9605	0.8264	1.5113	27.3416	895.513	True
5	1175	57.981	35.1354	1.6502	0.7955	0.6155	-1.5521	82.0417	147.133	False
6	718	47.2395	23.002	2.0537	0.8734	0.8022	-1.0816	93.7911	437.035	False
7	16316	190.924	121.711	1.5687	0.7705	0.7618	-1.2586	206.568	82.4135	True
8	2720	92.088	38.5852	2.3866	0.908	0.9547	-1.1428	215.941	986.257	True
9	506	33.0898	22.5781	1.4656	0.731	0.8447	-0.2309	223.868	412.781	False
10	6078	103.767	76.7807	1.3515	0.6727	0.9519	-0.9596	258.575	697.738	False
11	5138	135.156	50.3775	2.6829	0.9279	0.8874	-0.8946	375.376	523.812	False
12	749	51.0208	20.659	2.4697	0.9144	0.8874	1.1423	366.493	18.0734	True
13	2951	92.7411	57.1766	1.622	0.7873	0.6519	0.6626	412.099	797.415	False
14	3683	112.658	46.4293	2.4264	0.9111	0.7914	-0.9592	433.344	623.77	False
15	623	33.6165	26.9149	1.249	0.5991	0.8534	-1.4105	448.144	1010.89	True
16	3483	74.1218	60.6446	1.2222	0.575	0.9286	-0.204	483.984	66.1174	False
17	3784	85.9898	57.0703	1.5067	0.748	0.9309	-0.1766	493.888	417.965	False
18	4829	127.96	50.7336	2.5222	0.918	0.7542	-1.2747	583.146	228.783	False
19	2220	83.1379	43.4719	1.9125	0.8524	0.7754	0.0297	577.841	853.017	False
20	4318	144.296	45.1555	3.1955	0.9498	0.6869	1.4075	580.847	622.689	False
21	669	36.2679	35.241	1.0291	0.2363	0.673	1.4264	577.549	741.005	False
22	2037	75.2328	35.6451	2.1106	0.8806	0.9122	-0.2022	721.922	650.957	False
23	1103	54.1002	27.7589	1.9489	0.8583	0.9026	-0.7329	709.348	70.3019	False
24	812	45.0748	28.6241	1.5747	0.7725	0.7785	0.6665	738.431	13.2463	True

25	1992	69.0062	38.1155	1.8104	0.8336	0.9009	-1.2002	817.186	92.8996	False
26	1044	45.928	31.6235	1.4523	0.7252	0.8509	0.2821	817.663	289.875	False
27	38322	496.242	167.137	2.9691	0.9416	0.5887	-1.5553	972.902	712.084	True
28	2059	64.4016	53.6924	1.1995	0.5522	0.708	-1.4007	899.013	923.481	False
29	569	32.4583	24.6996	1.3141	0.6488	0.8493	-0.8213	949.207	269.23	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	284	429	367.806	24	15.3253	7.4207
2	10	250	375	1596.95	79	20.2146	7.3007
3	8	259	583	157.196	10	15.7196	7.2057
4	3	184	418	134.125	6	22.3541	8.1476
5	9	234	275	210.368	17	12.3746	5.2902
6	2	193	414	68.4264	4	17.1066	8.307
7	26	775	1243	1680.62	80	21.0078	7.1998
8	3	55	210	61.7696	5	12.3539	6.4488
9	6	303	696	669.955	34	19.7046	7.8869
10	6	278	662	261.309	13	20.1007	7.5827
11	13	318	660	1008.68	49	20.5853	7.8377
12	3	126	262	183.61	6	30.6017	6.8891
13	13	460	1038	1583.83	76	20.8399	7.061
14	11	334	463	330.593	21	15.7425	7.3598
15	4	130	342	352.049	14	25.1463	7.5961
16	5	172	447	347.309	19	18.2794	8.499
17	4	151	431	180.095	10	18.0095	8.0738
18	18	496	814	820.411	50	16.4082	7.4463
19	7	272	601	771.625	32	24.1133	7.6143
20	13	311	542	467.747	21	22.2737	7.8907
21	11	204	369	386.25	23	16.7935	6.8089
22	3	147	410	421.889	11	38.3536	8.8798
23	4	93	231	169.61	8	21.2013	7.4866
24	4	68	135	49.5563	5	9.9113	5.5824
25	6	190	379	585.73	24	24.4054	7.2164
26	3	108	403	278.321	9	30.9245	7.5782
27	20	593	1091	711.642	37	19.2336	8.1042
28	8	252	475	1126.76	54	20.866	9.0593
29	1	50	245	48.3553	2	24.1777	5.3905

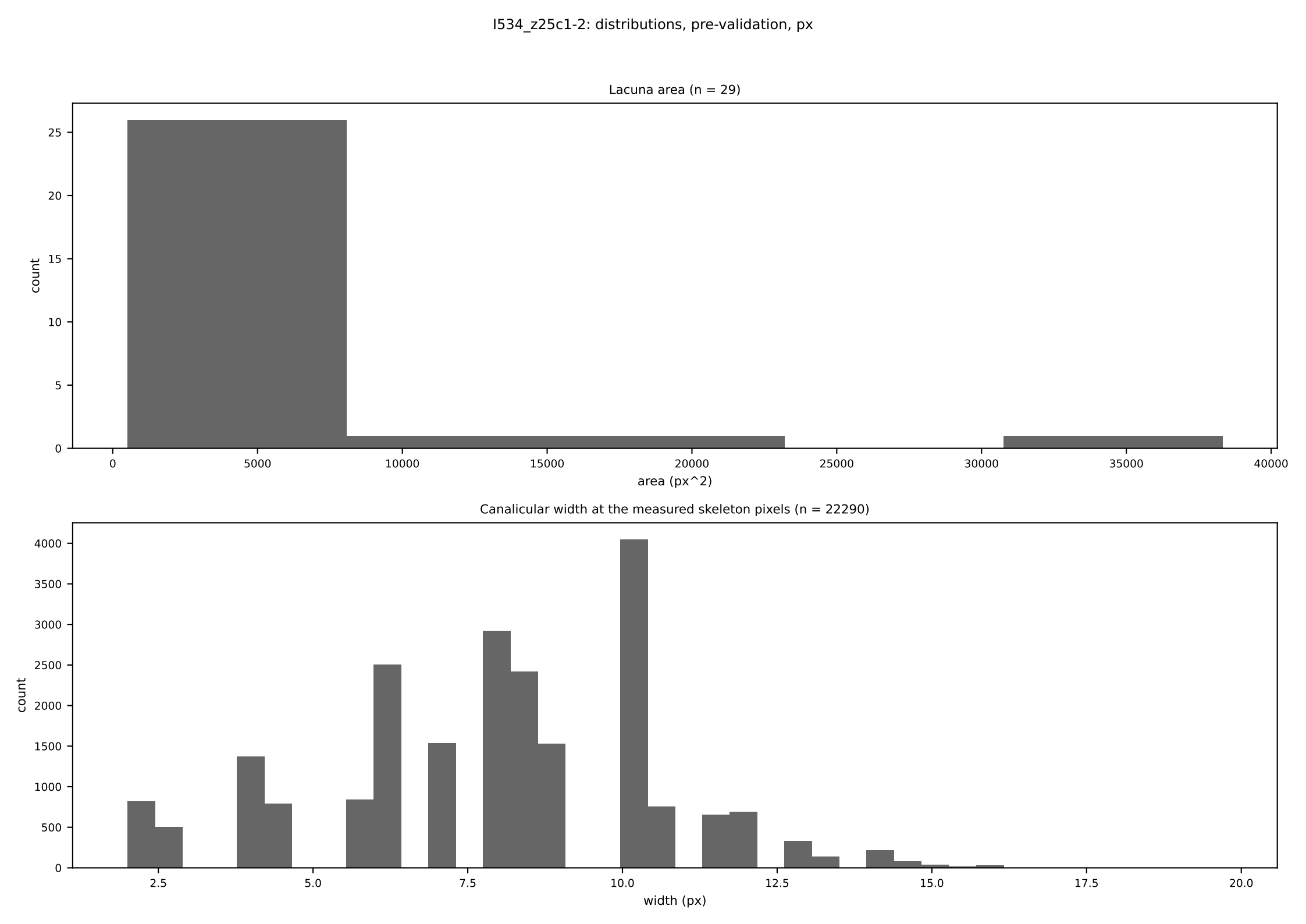
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	685.506	8694	14795	0.596	0.572	0.0326662	0.0289963	1.3129
2	192.167	4809	7186	0.6966	0.6322	0.0519859	0.0521848	5.2038
3	751.654	12022	23386	0.6838	0.6378	0.0215438	0.0249294	1.0643
4	629.754	9543	16652	0.6904	0.571	0.0192812	0.0251021	0.4764
5	206.965	6090	8616	1	0.9867	0.0384236	0.0319174	4.3486
6	132.125	5891	13464	1	1	0.0327618	0.0307487	1.5137
7	930.744	18415	31730	0.8227	0.7587	0.0420853	0.0391743	2.7935
8	237.965	7174	17084	0.7631	0.6986	0.00766657	0.0122922	1.2607
9	107.154	5665	16255	1	1	0.0534863	0.0428176	5.5994
10	334.475	11891	25749	1	1	0.023379	0.0257097	1.7939
11	366.517	12817	22622	1	1	0.0248108	0.0291751	3.5469
12	128.947	4467	8594	0.694	0.6341	0.0282068	0.0304864	2.3265
13	419.345	11919	24709	1	1	0.0385938	0.042009	3.1001
14	332.676	11820	20074	1	1	0.0282572	0.0230647	3.3065
15	114.184	3796	10383	0.6248	0.5844	0.0342466	0.0329385	3.5031

16	249.078	9533	20117	1	0.9067	0.0180426	0.02222	2.0074
17	262.877	9914	22390	1	1	0.015231	0.0192497	1.5216
18	405.588	13364	31075	1	1	0.0371146	0.0261947	4.438
19	278.09	9508	20761	1	1	0.0286075	0.0289485	2.5172
20	406.416	12535	23662	1	1	0.0248105	0.0229059	3.1987
21	166.752	5502	11604	1	1	0.0370774	0.0317994	6.5966
22	220.35	8451	20450	1	1	0.0173944	0.0200489	1.3615
23	153.238	5825	12202	1	0.9659	0.0159657	0.0189313	2.6103
24	188.368	3275	4996	0.6244	0.5829	0.0207634	0.0270216	2.1235
25	215.037	8340	19166	1	0.9972	0.0227818	0.0197746	2.7902
26	146.853	6827	18445	1	1	0.0158195	0.0218487	2.0429
27	2457.27	34561	57941	0.6713	0.6268	0.0171581	0.0188295	0.8139
28	316.676	7145	14641	1	1	0.0352694	0.0324431	2.5262
29	106.225	5740	16217	1	0.9941	0.0087108	0.0151076	0.9414

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	253	303	322.764	490.688	6	7	6
2	230	355	282.38	430.262	4	6	6
3	232	387	297.007	664.328	5	7	5
4	125	176	213.995	482.416	5	5	8
5	206	207	265.723	312.108	9	5	3
6	179	354	215.924	466.818	7	5	5
7	718	1033	869.411	1404.17	16	17	22
8	55	55	66.1838	236.995	2	1	0
9	294	650	344.522	803.796	6	9	9
10	224	373	303.238	737.742	7	6	8
11	296	584	370.019	751.855	9	7	11
12	114	152	140.154	300.764	3	2	6
13	416	877	530.345	1180.76	10	16	16
14	324	453	392.161	534.487	6	9	9
15	109	303	145.397	390.191	2	3	2
16	125	276	191.196	502.961	3	7	6
17	64	125	167.054	486.233	2	4	6
18	480	694	566.801	924.879	11	15	9
19	211	392	308.179	701.21	6	9	11
20	248	365	344.522	609.73	7	6	10
21	201	254	227.267	415.12	4	2	6
22	97	176	169.196	477.416	4	6	6
23	76	125	111.225	266.764	2	4	5
24	52	52	70.0416	151.711	1	0	3
25	182	297	215.338	424.363	6	4	6
26	105	217	127.782	467.073	4	3	3
27	409	578	652.186	1199.08	14	15	13
28	178	210	283.451	539.487	4	5	6
29	11	11	57.598	284.392	1	3	4


I534_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 29)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22290)
count
width (px)